A2 · Unit 4 · WPH14 Further Mechanics, Fields & Particles

Nuclear & Particle Physics

What is everything made of? A century ago the answer was "atoms"; then it was "electrons, protons and neutrons"; now it is quarks and leptons. Each step came from firing particles at other particles, harder and harder, and reading the tracks that came out. This page follows that story: alpha scattering, electron beams, accelerators, detectors, and the conservation laws that decide which particle reactions can happen.

Inside the atom: nucleon and proton numbers

An atom has a tiny, dense, positively charged **nucleus** made of protons and neutrons (together called **nucleons**), surrounded by electrons. The nucleus is about 10^{-15} to 10^{-14} m across, while the atom is about 10^{-10} m across: if the nucleus were a marble, the atom would be the size of a football stadium. Yet the nucleus holds more than 99.9% of the mass.

A nucleus is labelled with two numbers written in front of the chemical symbol, ${}^{A}_{Z}\mathrm{X}$:

DEFINITION

The **nucleon number** (mass number) A is the total number of protons and neutrons in the nucleus.

The **proton number** (atomic number) Z is the number of protons in the nucleus. It decides which element the atom is. The number of neutrons is $A - Z$.

For example, ${}^{14}_{6}\mathrm{C}$ has 6 protons and $14 - 6 = 8$ neutrons. Nuclei of the same element with different numbers of neutrons, such as ${}^{12}_{6}\mathrm{C}$ and ${}^{14}_{6}\mathrm{C}$, are **isotopes**: same chemistry, different mass. An alpha particle is a helium nucleus, ${}^{4}_{2}\mathrm{He}$: 2 protons and 2 neutrons, charge $+2e$.

Alpha scattering and the nuclear model

How the picture of the atom changed

- **Early 1800s:** Dalton pictures atoms as tiny, solid, indivisible spheres.
- **1897:** Thomson discovers the electron, so atoms must have smaller parts. His "plum pudding" model has electrons dotted through a sphere of spread-out positive charge.
- **1909–1911:** Geiger and Marsden fire alpha particles at thin gold foil; Rutherford explains the results with a tiny positive **nucleus**.
- **1913:** Bohr puts the electrons in fixed energy levels around the nucleus.
- **1919 and 1932:** the proton is identified, then Chadwick discovers the neutron.

- **1960s onwards:** high-energy electron scattering shows protons and neutrons are themselves made of **quarks**.

Each model was replaced because an experiment produced results it could not explain. That is how scientific models change over time.

The experiment

A beam of alpha particles from a radioactive source was aimed at a very thin gold foil in a vacuum (so the alphas were not stopped by air). A movable detector, a zinc sulfide screen that gives a flash of light when hit, counted the alphas scattered at each angle. The results:

- **Almost all** the alpha particles passed straight through or were deflected by only a few degrees.
- A **small fraction** were deflected through large angles.
- A **very few** (roughly 1 in 8000) bounced back through more than 90°.

KEY IDEA

What the results tell us

- **Most alphas pass straight through**, so most of the atom is **empty space**.
- **A few are deflected through large angles**, so there must be a very strong electric field somewhere. That needs a large charge concentrated in a **very small volume**: the nucleus.
- **Some bounce straight back.** An alpha particle is about 7000 times more massive than an electron, so only something much more massive than the alpha could turn it round. Nearly all the **mass** must be in the nucleus too.
- The alphas are **repelled**, so the nucleus is **positively charged**.

In the plum pudding model the positive charge is spread over the whole atom, so its field is weak everywhere and the electrons are far too light to turn an alpha round. It predicts only tiny deflections, which is why it had to go.

WORKED EXAMPLE

An alpha particle with kinetic energy 5.0 MeV is fired directly at a gold nucleus (Z = 79). Calculate its distance of closest approach. What does this tell you about the size of the gold nucleus?

Show solution

At the closest point the alpha stops for an instant, so all its kinetic energy has become electric potential energy: $E_k = \dfrac{Q_1 Q_2}{4\pi\varepsilon_0 r}$.

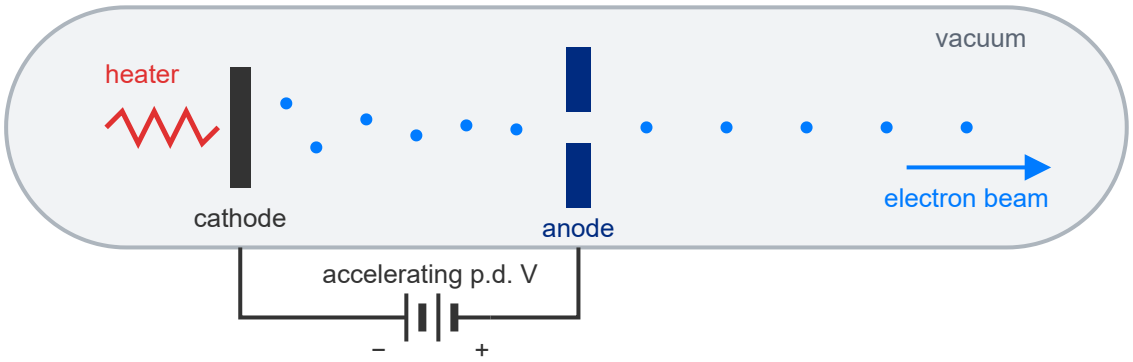
$E_k = 5.0 \times 1.60 \times 10^{-13} = 8.0 \times 10^{-13}$ J, $Q_1 = 79e$, $Q_2 = 2e$:

$$r = \frac{8.99 \times 10^{9} \times (79 \times 1.60 \times 10^{-19}) \times (2 \times 1.60 \times 10^{-19})}{8.0 \times 10^{-13}} = 4.5 \times 10^{-14}\ \text{m}$$

The alpha never touched the nucleus, so the radius of a gold nucleus must be **less than** about 4.5×10^{-14} m. (It is actually about 7×10^{-15} m.)

Electron beams: thermionic emission

To probe matter with electrons you first need a beam of them. Heat a metal filament (or a cathode next to it) to a high temperature and some of its free electrons gain enough energy to escape from the surface. This is **thermionic emission**. In a vacuum, the electrons are then accelerated towards a positive **anode**; some pass through a hole in it and form a narrow beam. The whole device is an **electron gun**.



An electron gun. The anode is positive relative to the cathode, so the electric field between them accelerates the emitted electrons. Beyond the anode there is no field, so they travel at constant velocity.

Each electron is accelerated through a p.d. V, so the electric field does work eV on it, which becomes kinetic energy. An electron accelerated through 1 V gains 1 electronvolt, $1\ \text{eV} = 1.60 \times 10^{-19}$ J.

$$eV = \tfrac{1}{2}mv^2 \quad \Rightarrow \quad v = \sqrt{\frac{2eV}{m}}$$

Once it has a beam, an **electric field** can deflect it (the force eE acts opposite to the field lines for an electron, so the beam bends in a parabola, like a projectile), and a **magnetic field** can bend it into a circular arc (the force Bev is always at right angles to the motion). Both ideas appear again in accelerators and detectors.

WORKED EXAMPLE

Electrons are emitted from a hot cathode and accelerated through a p.d. of 2.0 kV. Calculate their kinetic energy in joules and their speed.

Show solution

$E_k = eV = 1.60 \times 10^{-19} \times 2.0 \times 10^{3} = 3.2 \times 10^{-16}$ J (that is 2.0 keV)

$v = \sqrt{\dfrac{2E_k}{m}} = \sqrt{\dfrac{2 \times 3.2 \times 10^{-16}}{9.11 \times 10^{-31}}} = 2.7 \times 10^{7}\ \text{m s}^{-1}$, about 9% of the speed of light. At much higher p.d.s this simple equation stops working because the speed cannot exceed c.

Particle accelerators: linac and cyclotron

To smash particles together hard enough to see inside nucleons or create new particles, they need energies of GeV or more. A single accelerating gap would need billions of volts, which is impossible, so accelerators give the particle many smaller pushes. **Electric fields** do the accelerating; **magnetic fields** steer.

The linear accelerator (linac)

A linac is a long line of metal **drift tubes** in a vacuum, connected alternately to the two terminals of a high-frequency alternating supply. Inside a tube there is no electric field (the metal shields it), so the particle drifts at constant speed. In each **gap** between tubes there is a strong field, and the supply is timed so that the field always points forwards when the particle is crossing a gap. While the particle is hidden inside a tube, the supply reverses, ready for the next gap. Each gap adds energy QV.

The particle gets faster at every gap, but it must spend the same time (half a cycle of the supply) inside each tube. So the tubes must get **longer** along the accelerator.

The cyclotron

A cyclotron curls the path up so the same gap can be used again and again. Two hollow D-shaped electrodes, the **dees**, sit in a vacuum between the poles of a large magnet, with a gap between them and an alternating p.d. across the gap. Particles start near the centre.

- Inside a dee there is no electric field. The **magnetic field** makes the particle move in a **semicircle** (the magnetic force does no work, so the speed is constant).
- Each time the particle reaches the gap, the **electric field** across it accelerates the particle into the other dee. The p.d. has reversed in the meantime, so it pushes forwards both ways.
- Faster particles move in bigger circles ($r = p/BQ$), so the path spirals outwards until the particle reaches the edge and is extracted.

Why the cyclotron frequency doesn't depend on speed

DERIVATION NOT REQUIRED · YOU ONLY NEED TO USE THE RESULT

1. Inside a dee the particle moves in a semicircle of radius r at speed v. The time to go half way round is:

$$t_{1/2} = \frac{\pi r}{v}$$

2. The radius is $r = mv/BQ$ (derived in the next section). Substitute:

$$t_{1/2} = \frac{\pi}{v} \cdot \frac{mv}{BQ} = \frac{\pi m}{BQ}$$

The speed cancels. A faster particle goes round a bigger circle, and the two effects exactly balance.

3. One cycle of the supply must last two half-turns, so the period is $2\pi m/BQ$ and the frequency is:

$$\boxed{f = \frac{BQ}{2\pi m}}$$

So a fixed-frequency supply stays in step with the particle on every turn, from the centre to the edge.

WORKED EXAMPLE

A cyclotron for protons has dees of radius 0.50 m in a magnetic field of 1.5 T. Calculate (a) the frequency of the alternating supply and (b) the maximum kinetic energy of the protons, in MeV.

Show solution

(a) $f = \dfrac{BQ}{2\pi m} = \dfrac{1.5 \times 1.60 \times 10^{-19}}{2\pi \times 1.67 \times 10^{-27}} = 2.3 \times 10^{7}$ Hz (23 MHz)

(b) The largest orbit has $r = 0.50$ m, so the maximum momentum is $p = BQr = 1.5 \times 1.60 \times 10^{-19} \times 0.50 = 1.2 \times 10^{-19}$ kg m s^{-1}.

$$E_k = \frac{p^2}{2m} = \frac{(1.2 \times 10^{-19})^2}{2 \times 1.67 \times 10^{-27}} = 4.3 \times 10^{-12}\ \text{J} = \frac{4.31 \times 10^{-12}}{1.60 \times 10^{-13}}\ \text{MeV} = 27\ \text{MeV}$$

Detecting particles and r = p/BQ

Particles are far too small to see, so detectors rely on two effects.

- **Ionisation.** A fast charged particle knocks electrons out of the atoms it passes, leaving a trail of ions. In a cloud chamber the ions trigger droplets, in a bubble chamber tiny bubbles, and in modern detectors an electrical signal. Uncharged particles (neutrons, neutrinos, photons) hardly ionise, so they leave **no track**: you spot them from gaps and from where new tracks suddenly begin.
- **Deflection.** The detector sits in a magnetic field, so charged tracks curve. The **direction** of curving gives the sign of the charge, and the **radius** gives the momentum.

Deriving r = p/BQ

SPEC: YOU MUST BE ABLE TO DERIVE THIS

1. A particle of charge Q moves at speed v at right angles to a uniform field B. The magnetic force on it is:

$$F = BQv$$

This force is always perpendicular to the velocity, so it provides the centripetal force and the particle moves in a circle.

2. Set the magnetic force equal to the centripetal force:

$$BQv = \frac{mv^2}{r}$$

3. Cancel one factor of v and rearrange for r:

$$r = \frac{mv}{BQ}$$

4. mv is the momentum p, so:

$$\boxed{r = \frac{p}{BQ}}$$

A larger momentum gives a larger, less curved track; a stronger field or a larger charge curls it up more tightly.

Charged particle in a magnetic field

FORMULA SHEET

$$r = \frac{p}{BQ}$$

r: radius of path / m p: momentum / kg m s^{-1} B: flux density / T Q: charge / C

KEY IDEA

Reading particle tracks

- **Which way it curves** tells you the sign of the charge (use Fleming's left-hand rule, remembering that the current is opposite to the motion for a negative particle). Opposite charges curve opposite ways.
- **How tightly it curves** tells you the momentum: $p = BQr$. A spiral means the particle is losing momentum as it goes.
- **Thick, dense tracks** come from slow or heavily charged particles, which ionise strongly; fast light particles leave faint tracks.
- **Where tracks start or stop** without anything visible, an uncharged particle has arrived or left. Conservation of charge and momentum at that point tells you about the invisible particle.

WORKED EXAMPLE

A proton leaves a circular track of radius 0.25 m in a bubble chamber with a magnetic field of 1.2 T. Calculate its momentum, in SI units and in MeV/c, and its kinetic energy in MeV.

Show solution

$p = BQr = 1.2 \times 1.60 \times 10^{-19} \times 0.25 = 4.8 \times 10^{-20}\ \text{kg m s}^{-1}$

In MeV/c: $pc = 4.8 \times 10^{-20} \times 3.00 \times 10^{8} = 1.44 \times 10^{-11}\ \text{J} = 90\ \text{MeV}$, so $p = 90\ \text{MeV}/c$.

$E_k = \dfrac{p^2}{2m} = \dfrac{(4.8 \times 10^{-20})^2}{2 \times 1.67 \times 10^{-27}} = 6.9 \times 10^{-13}\ \text{J} = 4.3\ \text{MeV}$. Much less than its rest energy (938 MeV), so the non-relativistic formula is fine.

High energies, E = c²Δm and particle units

Why high energies are needed

There are two reasons. First, to **see** small detail you need a probe whose de Broglie wavelength $\lambda = h/p$ is about as small as the detail. A nucleon is about 10^{-15} m across, so the probe needs $p \approx h/\lambda = 6.6 \times 10^{-19}\ \text{kg m s}^{-1}$, which for an electron means an energy of about $pc \approx 2 \times 10^{-10}\ \text{J}$, roughly 1 GeV. Lower energies just diffract round the nucleon without resolving its insides.

Second, to **create** new particles you need enough energy to make their mass. Energy and mass are equivalent:

Mass–energy

FORMULA SHEET

$$\Delta E = c^2 \Delta m$$

ΔE: energy / J Δm: mass / kg $c = 3.00 \times 10^8\ \text{m s}^{-1}$

In a high-energy collision, kinetic energy can turn into the mass of brand new particles. To discover a very massive particle, like the top quark, you need a collision with at least that much energy to spare.

Creation and annihilation

Energy can only become matter in **particle–antiparticle pairs**, because charge and the other quantum numbers must be conserved. In **pair production** a high-energy photon passing close to a nucleus becomes an electron and a positron. The photon needs at least enough energy for both rest masses, $2m_e c^2 = 1.02$ MeV; any extra becomes their kinetic energy. The nearby nucleus is needed to take up some momentum.

The reverse is **annihilation**: a particle meets its antiparticle and both vanish, their total energy reappearing as photons. An electron and positron at rest produce **two** photons of 0.511 MeV each, going in opposite directions so that the total momentum stays zero. PET scanners in hospitals detect exactly these photon pairs.

MeV, GeV and MeV/c²

Joules are inconveniently large for single particles, so particle physicists use electronvolts. Because $E = mc^2$, a mass can be given as an energy divided by c^2, and momentum as an energy divided by c.

Unit	Measures	In SI units
1 MeV	energy	$1.60 \times 10^{-13}\ \text{J}$
1 GeV	energy	$1.60 \times 10^{-10}\ \text{J}$
1 MeV/c²	mass	$\dfrac{1.60 \times 10^{-13}}{(3.00 \times 10^8)^2} = 1.78 \times 10^{-30}\ \text{kg}$
1 GeV/c²	mass	$1.78 \times 10^{-27}\ \text{kg}$
1 MeV/c	momentum	$5.33 \times 10^{-22}\ \text{kg m s}^{-1}$

Useful values: electron 0.511 MeV/c², proton 938 MeV/c², neutron 940 MeV/c². So a proton's rest mass is just under 1 GeV/c².

WORKED EXAMPLE

(a) Calculate the minimum photon energy, in MeV and in J, needed to produce an electron–positron pair, and the wavelength of such a photon. (b) A proton and an antiproton, both at rest, annihilate into two identical photons. Calculate the energy of each photon in GeV.

Show solution

(a) $E = 2m_e c^2 = 2 \times 9.11 \times 10^{-31} \times (3.00 \times 10^8)^2 = 1.64 \times 10^{-13}\ \text{J} = 1.02\ \text{MeV}$.

$\lambda = \dfrac{hc}{E} = \dfrac{6.63 \times 10^{-34} \times 3.00 \times 10^8}{1.64 \times 10^{-13}} = 1.2 \times 10^{-12}\ \text{m}$, a gamma ray.

(b) The total rest energy is $2 \times 938\ \text{MeV} = 1876\ \text{MeV}$, shared between two photons, so each has $938\ \text{MeV} = 0.938\ \text{GeV}$.

Longer lifetimes at high speed

Many particles are unstable and decay after a characteristic lifetime. When they travel at a speed close to the speed of light, their lifetime measured in the laboratory is **longer** than when they are at rest, so they travel much further before decaying. This relativistic effect is negligible at everyday speeds and only becomes significant when the speed is a large fraction of c. Two examples: muons made by cosmic rays 10–15 km up in the atmosphere reach the ground, although at rest they live only 2.2 μs (about 660 m of travel even at the speed of light); and unstable particles made in accelerators leave longer tracks than you would expect from their rest lifetimes.

The standard model: quarks and leptons

The hundreds of particles found in accelerators fit a simple pattern. All matter particles are either **quarks** or **leptons**, which as far as we know are **fundamental** (not made of anything smaller).

Quarks	Charge / e	Baryon number	Leptons	Charge / e	Lepton number
up u, charm c, top t	$+\frac{2}{3}$	$+\frac{1}{3}$	electron e^-, muon μ^-, tau τ^-	−1	+1
down d, strange s, bottom b	$-\frac{1}{3}$	$+\frac{1}{3}$	neutrinos ν_e, ν_μ, ν_τ	0	+1

Quarks are never found on their own. They combine into particles called **hadrons**, in only two ways:

- **Baryons** are made of **three quarks** (qqq), for example the proton (uud: $\frac{2}{3} + \frac{2}{3} - \frac{1}{3} = +1$) and the neutron (udd: $\frac{2}{3} - \frac{1}{3} - \frac{1}{3} = 0$). Every baryon has baryon number +1.
- **Mesons** are made of a **quark and an antiquark** (q $\bar{q}$), for example the pions: π^+ = u $\bar{d}$, π^- = d $\bar{u}$ and π^0 (a mixture of u $\bar{u}$ and d $\bar{d}$). Mesons have baryon number 0.

The other two classes are **leptons**, the electron and its relatives, which are fundamental and are not made of quarks; and **photons**, the particles of electromagnetic radiation.

Antiparticles

Every particle has an **antiparticle** with the **same mass** and the **opposite** charge, baryon number and lepton number. The positron e^+ has charge +1 and lepton number −1. The antiproton $\bar{p} = \bar{u}\,\bar{u}\,\bar{d}$ has charge −1 and baryon number −1. The antineutron $\bar{n} = \bar{u}\,\bar{d}\,\bar{d}$ is still uncharged, but it is not the same as a neutron, because its baryon number is −1. A few neutral particles, such as the photon and the π^0, are their own antiparticles.

A pattern that made a prediction

Quarks and leptons come in matching pairs, called generations: (u, d) with (e, ν_e); (c, s) with (μ, ν_μ); and (t, b) with (τ, ν_τ). By 1977 the tau lepton and the bottom quark had been found, but the bottom quark had no partner. The symmetry of the model demanded one, the **top quark**, with charge $+\frac{2}{3}$. It was so massive (about 173 GeV/c², nearly the mass of a gold atom) that it took until 1995, and a much more powerful accelerator, to make it. Its discovery was a big success for the model.

Conservation laws and particle equations

Particle reactions are written like chemical equations. In every reaction certain quantities must be the same before and after. If any one of them is not conserved, the reaction **cannot happen**.

KEY IDEA

Always conserved

- **Charge** Q
- **Baryon number** B: +1 for baryons, −1 for antibaryons, 0 for everything else (+⅓ per quark)
- **Lepton number** L: +1 for leptons, −1 for antileptons, 0 for everything else
- **Energy** (including rest energy) and **momentum**

To check a reaction, make a small table with a column for each particle and rows for Q, B and L, then compare the totals on each side.

For example, in beta-minus decay a neutron turns into a proton:

$$\mathrm{n} \rightarrow \mathrm{p} + \mathrm{e}^- + \bar{\nu}_e$$

	n	→	p	e^-	$\bar{\nu}_e$	Conserved?
Charge Q	0		+1	−1	0	0 = 0 ✓

Baryon number B	+1	+1	0	0	1 = 1 ✓
Lepton number L	0	0	+1	−1	0 = 0 ✓

Without the antineutrino, lepton number would go from 0 to +1, so the antineutrino **must** be there. (Its existence was first suggested because the electrons from beta decay had a spread of energies, which only made sense if an unseen particle was carrying some away.) At the quark level, one down quark has become an up quark: $\mathrm{d} \rightarrow \mathrm{u} + \mathrm{e}^{-} + \bar{\nu}_{e}$.

EXAM TIP

Lepton number catches most people out. Remember that an **anti**neutrino and a **positron** both have $L = -1$. And don't forget energy: a single particle at rest can only decay into particles whose total rest mass is **smaller** than its own. That is why a free neutron can decay into a proton, but a free proton (lighter than a neutron) cannot decay into a neutron.

WORKED EXAMPLE

A negative pion collides with a stationary proton: $\pi^{-} + \mathrm{p} \rightarrow \mathrm{n} + \mathrm{X}$. Use conservation laws to identify X. Then explain why the reaction $\mathrm{p} \rightarrow \mathrm{n} + \pi^{+}$ cannot happen for a free proton.

Show solution

Charge: $-1 + 1 = 0 + Q_X$, so $Q_X = 0$. Baryon number: $0 + 1 = 1 + B_X$, so $B_X = 0$. Lepton number: $0 = 0 + L_X$, so $L_X = 0$.

X is uncharged with $B = 0$ and $L = 0$: a neutral meson such as the π⁰ (or a photon).

For $\mathrm{p} \rightarrow \mathrm{n} + \pi^{+}$: charge (+1 = 0 + 1), baryon number (1 = 1 + 0) and lepton number (0 = 0) are all conserved. But the products have a total rest energy of $940 + 140 = 1080$ MeV, more than the proton's 938 MeV. A particle at rest cannot decay into heavier products, so energy conservation forbids it.

Quick checks

QUICK CHECK

In the alpha-scattering experiment, what was the evidence that the nucleus contains most of the atom's mass?

- Most alpha particles passed straight through the foil.
- A few alpha particles were scattered back through more than 90°.
- The alpha particles were repelled, not attracted.
- The foil had to be very thin.

Only something much more massive than the alpha particle could turn it right round. Passing straight through shows the atom is mostly empty space; being repelled shows the nucleus is positive.

QUICK CHECK

A proton and an alpha particle move at the same speed at right angles to the same uniform magnetic field. What is the ratio (radius of alpha's path) / (radius of proton's path)?

- 1/2
- 1
- 2
- 4

$r = mv/BQ$. The alpha has 4 times the mass but twice the charge, so its radius is $4/2 = 2$ times as big.

QUICK CHECK

Why do the drift tubes in a linear accelerator get longer along its length?

- To give a larger accelerating p.d. at each gap
- The particle speeds up, but must spend the same time (half a cycle) inside each tube
- Because the particle's mass increases, so it needs more room
- To reduce the magnetic field needed to steer the beam

The supply frequency is fixed, so the particle must arrive at each gap exactly half a cycle after the last one. Faster particles cover more distance in that time, so each tube must be longer.

QUICK CHECK

Which of these is a meson?

- u u d
- u $\bar{d}$
- $\bar{u}$ $\bar{u}$ $\bar{d}$
- u d

A meson is a quark plus an antiquark: u $\bar{d}$ is the π^+. uud is a proton (baryon), $\bar{u}$ $\bar{u}$ $\bar{d}$ an antiproton (antibaryon), and u d is not allowed (charge +⅓).

QUICK CHECK

Which property of the electron antineutrino $\bar{\nu}_e$ is correct?

- Charge −1, lepton number +1
- Charge 0, lepton number +1
- Charge 0, lepton number −1
- Charge +1, lepton number −1

An antiparticle has the opposite charge and lepton number of its particle. The neutrino has charge 0 and $L = +1$, so the antineutrino has charge 0 and $L = -1$.

QUICK CHECK

Which reaction is impossible because baryon number is not conserved?

- $\pi^+ \rightarrow \mu^+ + \nu_\mu$
- $p + p \rightarrow p + p + \pi^0$
- $p \rightarrow \pi^+ + \pi^0$
- $e^- + e^+ \rightarrow \gamma + \gamma$

The proton has $B = +1$ but pions are mesons with $B = 0$, so B would go from 1 to 0. Charge is conserved (+1 = +1 + 0), but one failure is enough.